

Programming Using C++ Level 1

While Loops



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Definition

◆ A while loop is a control structure used to repeat a block of code as long as a specified condition is true.

Explanation

The while loop checks the condition before each iteration. If the condition is true, the code inside the loop executes. After that, the condition is checked again. In this program, i starts at 1 and increases by 1 after each iteration. The loop continues while $i \leq N$.

Steps in main()

1. Read a number from the user using ReadInput().
2. Store the input in N.
3. Set $i = 1$.
4. Check if $i \leq N$.
5. Print i if the condition is true.
6. Increase i by 1.
7. Repeat until $i \leq N$ becomes false.

💡 Example

If the user enters 5:

1
2
3
4
5

➡ Result

The program prints all numbers from 1 to N, each on a separate line.

Note

◆ The statement `i++` is important because it changes the loop variable and eventually makes the condition false.

⊘ Warning

- ◆ If the condition never becomes false, the while loop can become an infinite loop.



Comparison

while loop

- ✓ Checks the condition before each iteration.
- ✓ Useful when the number of repetitions depends on a condition.
- ✓ The loop variable must be updated manually.

for loop

- ✓ Usually convenient when the number of repetitions is known.
- ✓ Initialization, condition, and update are written together.
- ✓ In this example, a for loop can produce the same result.

✦✦ Summary

- ✓ A while loop repeats code while its condition is true.
- ✓ The condition is checked before every iteration.
- ✓ Always make sure the loop can eventually terminate.

Thank you 😊